A-A-51140A
23 March 1990
SUPERSEDING
A-A-51140
14 January 1985

COMMERCIAL ITEM DESCRIPTION

CHUCKS, DRILL, KEY TYPE, THREE-JAW, GEARED

The General Services Administration has authorized the use of this commercial item description.

Abstract.

This description covers geared type three-jaw drill chucks capable of being adjusted to hold any size straight shank tool within the capacity of the chuck. The jaws of the chuck shall be opened and closed by means of a pinion-type key.

Classification. The drill chucks shall be of the following classes, styles, and sizes. The class, style, and size to be furnished shall be as specified, (see tables I, II, or III).

Class 1 - Tapered-mount

Style A - Medium duty chucks - Intended for use on stationary and portable power tools.
Style B - Heavy duty chucks - Suitable for use with heavy duty portable power tools.
Style C - Extra heavy duty chucks - Shall be used with production drilling machines, jig borers, radial drill presses, milling machines and lathes.

Class 2 - Threaded-mount

Style A - Medium duty chucks - Intended for use on stationary and portable power tools.
Style B - Heavy duty chucks - Suitable for use with heavy duty portable power tools.

Beneficial comments recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd., Memphis TN 38114-5051.

AMSC N/A FSC 3460

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The machine shall have the following salient characteristics:

1. Material. The chuck body shall be constructed of carbon steel or tool steel. All chuck jaws shall be made of alloy steel. Chuck keys shall be made of cold rolled steel or equivalent.

2. Hardness. The sides of the keyholes shall be hardened to not less than Rockwell 48C. The gripping surface of each chuck jaw shall have a Rockwell C scale hardness value of not less than 52. Each chuck key shall be hardened on the bearing surface and shank to a Rockwell C scale value of not less than 42.

3. Tapered-mount hole chuck. The tapered-mount hole of class 1 chucks shall conform to the requirements of tables I, II or III for the specific chuck size specified.

4. Threaded-mount hole chuck. The threaded-mount hole of class 2 chucks shall conform to the dimensions of tables I or II for the chuck size specified. The threads shall conform to class 2B tolerance requirements of FED-STD-H28. The direction of the threads shall be such that to remove the chuck from a tool spindle, the chuck must be turned in the same direction the spindle rotates when in use. Chucks designed for use on reversible tools shall have a retaining screw, or other suitable means to secure the chuck during reverse operations.

5. Bearings. All style A and style B drill chucks shall have plain bearings. Style C drill chucks shall contain ball thrust bearings. The ball thrust bearings shall be of a standard, commercial quality, readily available for replacement purposes. If the thrust bearings are not "sealed-for-life" means shall be provided for lubrication.

6. Drill chuck capacity. The drill chuck shall conform to the drill holding capacity listed in the respective tables for the class and style chuck required. Each drill chuck shall be plainly marked with the manufacturer's name, model number and capacity.

7. Chuck key. Unless otherwise specified in the contract or order, each key operated chuck shall be furnished with a standard pinion type key. When specified in the contract or order, a self-ejecting safety key shall be furnished. The pinion key shall mesh with and operate smoothly with the geared portion of the chuck for 360° of rotation. Each key handle shall have sufficient diameter and leverage to lock the tool firmly in the chuck without bending or breaking the handle. The handle may be a steel pin either rounded on both ends or flattened at one end for application of additional thumb pressure. Self-ejecting keys shall have a built in mechanism which shall eject the key from the chuck body when hand pressure upon the key is relieved.

8. Workmanship. Each drill chuck and key shall be free from any imperfections or defects which may impair serviceability, durability or appearance.

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TABLE I. Sizes and capacities for classes 1 and 2, style A chucks.

Size	Capacity (inches)	Mount hole end of chuck: Jacobs taper number class 1, style A	Mount hole end of chuck: Nominal dia. (inches), and thread, class 2, style A	Chuck dimensions, inches: Overall length closed	Chuck dimensions, inches: Sleeve dia.
1	0-5/32	0	5/16 x 24	1-3/8	53/64
2	1/64-5/32	0	--	1-45/64	1
3	0-1/4	---	5/16 x 24	1-15/16	1-3/16
4	0-1/4	1	3/8 x 24	1-15/16	1-1/8
5	0-1/4	---	1/2 x 20	2-1/4	1-5/16
6	1/64-1/4	1	*3/8 x 24	2-1/16	1-7/64
7	1/64-1/4	---	1/2 x 20	2-23/64	1-9/32
8	1/64-5/16	1	*3/8 x 24	2-19/64	1-9/32
9	1/64-5/16	---	1/2 x 20	2-23/64	1-9/32
10	0-3/8	2	*3/8 x 24	2-3/4	1-11/16
11	0-3/8	---	1/2 x 20	2-3/4	1-25/32
12	1/64-3/8	---	3/8 x 24	2-23/64	1-19/64
13	1/64-3/8	---	1/2 x 20	2-23/64	1-19/64
14	1/32-3/8	2	*3/8 x 24	2-7/8	1-5/8
15	1/32-3/8	33	1/2 x 20	2-7/8	1-5/8
16	1/16-3/8	---	3/8 x 24	2-13/32	1-27/64
17	1/16-3/8	---	*1/2 x 20	2-13/32	1-27/64
18	0-1/2	2	1/2 x 20	3-7/16	1-25/32
19	0-1/2	33	5/8 x 16	3-5/64	1-25/32
20	0-1/2	6	---	3-7/16	2-5/64
21	1/32-1/2	2	1/2 x 20	3-3/8	2
22	1/32-1/2	6	5/8 x 16	3-3/8	2
23	1/32-1/2	33	---	3-3/8	2
24	1/16-1/2	---	*3/8 x 24	2-7/8	1-5/8
25	1/16-1/2	---	*1/2 x 20	2-7/8	1-5/8
26	5/64-1/2	33	3/8 x 24	3-3/32	1-25/32
27	5/64-1/2	---	*1/2 x 20	3-3/32	1-13/16
28	5/64-1/2	---	*5/8 x 16	3-3/32	1-13/16
29	1/32-5/8	3	5/8 x 16	3-55/64	2-7/32
30	1/8-5/8	3	5/8 x 16	3-13/16	2-19/64
31	1/8-5/8	---	3/4 x 16	3-13/16	2-3/8
32	5/32-3/4	3	---	3-55/64	2-7/32
33	3/16-3/4	3	5/8 x 16	3-13/16	2-3/8
34	3/16-3/4	4	3/4 x 16	3-57/64	2-3/8
35	1/4-13/16	3	---	3-21/64	2-33/64
36	3/16-1-1/32	5	---	5-25/32	3-7/64

*For use on reversible tools.

NOTE: All dimensions shall be not less than those shown. Where a range is shown, the dimensions shall be not greater than the lowest figure to not less than the highest figure.

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TABLE II. Sizes and capacities for classes 1 and 2, style B chucks.

Size	Capacity (inches)	Mount hole end of chuck		Chuck dimensions, inches	
		Jacobs taper number class 1, style B	Nominal dia. (inches), and thread, class 2, style B	Overall length fully closed	Sleeve dia.
1	0-1/4	1	*3/8 x 24	2-7/32	1-19/64
2	0-1/4	2 short	1/2 x 20	2-7/32	1-19/64
3	0-5/16	2 short	3/8 x 24	2-5/16	1-27/64
4	0-5/16	---	1/2 x 20	2-13/32	1-27/64
5	0-3/8	2	*1/2 x 20	2-3/4	1-25/32
6	0-3/8	---	5/8 x 16	2-3/4	1-25/32
7	0-1/2	6	5/8 x 16	3-7/16	2-1/16
8	0-1/2	---	45/64 x 16	3-17/32	2-1/16
9	0-1/2	---	3/4 x 16	3-17/32	2-1/16
10	1/8-5/8	3	5/8 x 16	3-13/16	2-5/16
11	3/16-3/4	---	5/8 x 16	3-1/64	2-3/8
12	3/16-3/4	3	3/4 x 16	3-57/64	2-3/8
13	1/4-13/16	3	---	4-11/32	2-9/16

NOTE: All dimensions shall be not less than those shown. Where a range is shown, the dimensions shall be not greater than the lowest figure to not less than the highest figure.
*For use on reversible tools.

TABLE III. Sizes and capacities for class 1, style C chucks.

Size	Capacity inches	Jacobs taper, mount hole end of chuck, class 1, style C	Chuck dimensions, inches	
			Overall length jaws closed	Sleeve diameter
1	0-1/4	2 short	2-11/32	1-9/16
2	0-3/8	2	2-45/64	1-49/64
3	0-1/2	3	3-27/32	2-23/64
4	1/32-1/2	3	3-53/64	2-15/64
5	1/8-5/8	3	4-5/16	2-5/8
6	1/32-5/8	3	3-57/64	2-7/16
7	1/8-3/4	4	4-57/64	2-53/64
8	3/16-3/4	4	4-41/64	2-51/64
9	3/8-1	5	5-1/2	3-21/32
10	3/16-1-1/32	5	4-23/32	3-7/64

NOTE: All dimensions shall be not less than those shown. Where a range is shown, the dimensions shall be not greater than the lowest figure to not less than the highest figure.

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9. Chuck runout and accuracy. When mounted on a true arbor, the plain bearing chuck shall run concentric within 0.004 inch total indicator (TIR) reading on any outside diameter of the chuck body. The maximum chuck runout shall be 0.004 inch TIR when checked with two true test plugs alternately tightened within the jaws, one having a diameter equal to one half and the other having the maximum capacity of the chuck. The runout shall be checked 1-1/2 inches from the end of the chuck body, on chucks 3/8 inch capacity and larger. The runout shall be checked at 1 inch from the end of the chuck body on chucks 1/4 to 5/16 inch capacity, and 3/4 inch from the end of the chuck body on chucks having 5/32 inch capacity. The maximum runout for ball bearing chucks shall be 0.003 inch TIR when checked 1 inch from the chuck nose.

10. Markings. Each drill chuck shall be plainly marked with the manufacturer's name, model number and capacity.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request shall be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without any adversely affecting performance requirements or exposing suppliers' employees to unlawful hazards from the recovered materials.

Environmental protection. During the manufacture of the drill chucks, materials hazardous to the ecological system as prohibited by the Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

Quality assurance.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection, examination, and test requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government. The Government

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reserves the right to perform any of the inspections, examinations, or tests set forth in this description where such inspections, examinations, and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

Responsibility for compliance. All items shall meet all requirements of salient characteristics paragraphs. The inspection set forth in this description shall become a part of the contractors overall inspection system or quality program. The absence of any inspection requirements in the description shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

Sampling inspection. As a minimum sampling inspection, the supplier shall randomly select the number of samples in table IV that is applicable to the quantity of items being purchased. Each sample item shall be inspected to determine conformance with the examination and the inspection. Should any of the sample items fail to meet either the examination or inspection, the entire lot shall be considered defective and unsuitable for acceptance. A lot is defined as all items presented at one time for acceptance.

TABLE IV. Sampling plan.

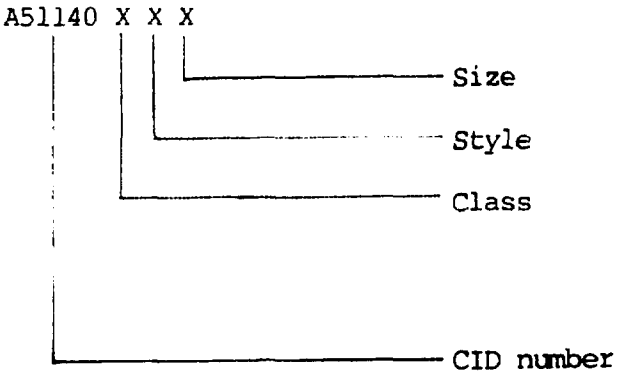
Quantity of items being purchased	Sample size	Quantity of items being purchased	Sample size
2 to 8	2	151 to 280	32
9 to 15	3	281 to 500	50
16 to 25	5	501 to 1,200	80
26 to 50	8	1,201 to 3,200	125
51 to 90	13	3,201 to 10,000	200
91 to 150	20		

Examination. Each drill chuck submitted for sample inspection shall be examined to determine conformance with all requirements of this specification.

Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be specified in the contract or order.

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CID based part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor. Where applicable Arabic numbers shall be substituted for Roman numerals.



Notes.

Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this commercial item description.

b. Class, style, size, and quantity required.

c. Packaging, if required specify level.

d. Specify either standard or safety type chuck keys, the quantity required, the manufacturer's name, catalog or other identifying numbers and types, model numbers and capacities of chuck for which the key is intended.

e. Other requirements.

Federal Standards FED-STD-376 and FED-STD-H28 are available from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.

MILITARY INTEREST:

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PREPARING ACTIVITY:

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Project 3460-0147

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-51140A	Chucks, Drill, Key Type, Three-Jaw, Geared

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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